

Lecture 10

Introduction to Ethics

Acknowledgements:

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Goal of Discussing Ethics

- **Not to lecture you on how to be a good person.**
 - But be a good person anyway.
- **Help you understand some of your ethical responsibilities.**
- **Help you avoid ethically challenging situations.**
- **Help you make decisions in ethically challenging situations.**
 - Help you understand your options and their ramifications.
- **Help you understand what other peoples' reactions might be to ethically challenging situations.**
 - What are typical responses/arguments you will face?
 - What are weakness/strengths of different arguments?

Warm-Up Exercise

**Who taught you the difference
between right and wrong?**





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Visual settings



Edit



Join by Web PollEv.com/davidwentzloff730



Who taught you the difference between right and wrong?



0

Parents		0%
Religion		0%
Government		0%
Friend / Mentor		0%
Other		0%

Scientific Expert



Director Nat. Inst. Allergy and Infectious Diseases
1984-2022

Pioneering work in HIV research and treatments

Presidential Medal of Freedom

National Medal of Science

M.D. Cornell + 53 honorary doctoral degrees

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Moral Expert?



Parents
Religion
Government
Friends
Teacher

Why Discuss Ethics?

...They are rules to live by so that society functions

- We live in a society where our actions affect the wellbeing of others and their actions affect us

...They are codified constraints on behavior

- Laws
 - Aristotle (350 B.C.): “The rule of law is better than the rule of any individual.”
- Professional Rules and Regulations
- Company Rules and Regulations

Ethics deals with moral choices made by individuals and organizations, especially when faced with conflicting principles

Science is Objective

- Scientific statements can be proven true or false



>



(True)

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>



(True)

$$2 + 2 = 5$$

(False)

Is Morality Objective?

- Can moral judgements be determined to be true or false?

It is wrong to steal

True/False?

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True/False?

One should not lie

True/False?

Others Are Not So Obvious

- Can moral judgements be determined to be true or false?

It is wrong to...

...modify human DNA

True/False?

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True/False?

...design facial recognition software

True/False?

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True/False?

...track online shopping behavior

True/False?

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- Can moral judgements be determined to be true or false?

It is wrong to...

...modify human DNA

True/False?

...design facial recognition software

True/False?

...track online shopping behavior

True/False?

...hack into someone else's phone

True/False?

Ethical Theories We'll Discuss

- 1. Utilitarianism**
- 2. Duty ethics**
- 3. Rights (contract) ethics**
- 4. Virtue ethics**

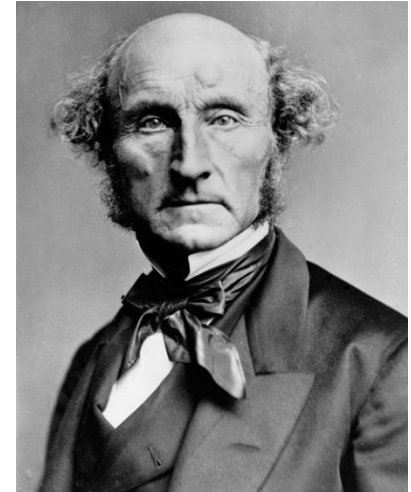
**For each theory, we'll define them and examine the pros and cons
You don't have to agree with any of these theories.**

Some other ethical theories we aren't discussing:

- Divine command theory: religion tells us what is right or wrong.**
- Subjective and cultural relativism: No universal right or wrong.**
- Ethical egoism – objectivism (Ayn Rand): Do what is in your own self interest.**

Utilitarianism (Consequentialism)

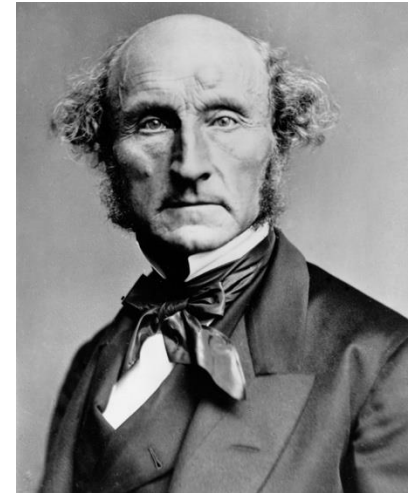
- Base a decision on a prediction of the outcome
- An action is good if its benefits exceed its harms (across society)
 - Denies the concept of morality
 - Attempts to be purely rational
- Promotes decisions “For the greater good...”
- Requires a cost-benefit analysis.



John Stuart Mill

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John Stuart Mill

Case For and Against

For

- **Focus on maximizing overall happiness.**
- **It is straightforward to apply (if you can do a cost benefit analysis).**
 - Removes conflicts over individual values, rights, ...

Against

- **It can be difficult to reduce everything to a number.**
 - E.g. How much is a life worth?
 - Ignores rights, such as privacy, property ownership, ...
- **Can be difficult to predict result of an action.**
- **It is not practical to make these calculations for every decision.**
- **Allows for selfish behavior**
 - This will make me happier than it will make you sad.
- **Focuses on result, not intent.**

Well-Known Example: The Trolley Problem

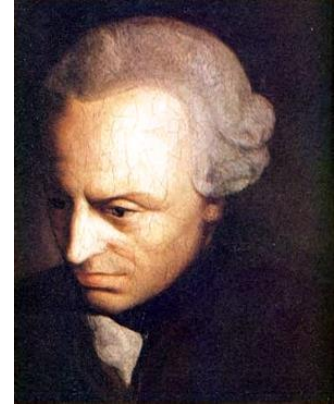


Many Dimensions Influence the Decision

- **How actively are you involved and what are your intentions?**
- **Your knowledge of the probability of events**
- **Which people involved**
 - **A criminal**
 - **A stranger**
 - **A friend**
 - **A relative**
 - **A famous/important person**
 - **Yourself**

Deontology: Duty Ethics

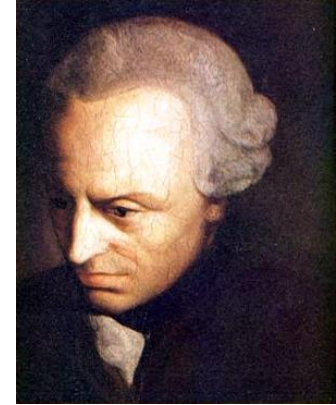
- **The morality of a decision depends on whether the action itself is right or wrong**
 - Do the right thing for the right reason because it is the right thing to do.
- **Example Duties**
 - Tell the truth, do good for others, gratitude, justice, self-improvement, non-maleficence (avoiding actions that do harm).



Immanuel Kant

Deontology: Duty Ethics

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Immanuel Kant



WikiLeaks



More on Deontology

- An action is good if the intent behind it is good.
 - Kant does not consider the consequences as important as the intent behind the action – counter-utilitarian
- Furthermore, an action is “good” if it can be universalized.
 - “What if everyone did that?” – would you want to live in this world

Case For and Against Duty Ethics

Case For

- Provides universal moral guidelines.
- Is based on intention instead of result.
 - Avoids the uncertainty of utilitarianism which examine the results of actions.
- Everyone is treated as moral equals.

Against

- Unclear what to do when rules conflict.
- Absolute in terms of intent instead of result
 - You can't lie to save your life
- Some “ethical” acts could make the world a worse place.

Deontology vs Utilitarianism





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Visual settings



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When poll is active respond at Pollev.com/davidwentzloff730



Straw Poll: To you lean more towards Utilitarianism or Deontology ethics?

0

0%

Utilitarianism

0%

Deontology

Social Contract / Rights-based

There are basic rights, such as the right to life, liberty, property, privacy, pursuit of happiness, free expression, education

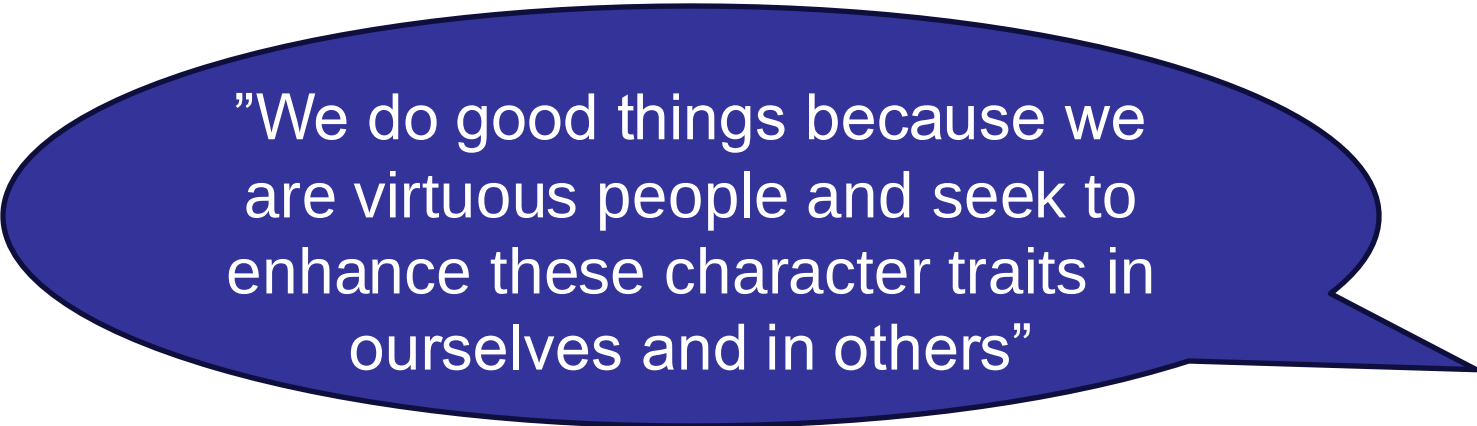
Absolute Rights	Guaranteed without exception
Limited Rights	May be restricted by other circumstances (interaction with other rights)
Negative Rights	You can pursue them if others don't interfere
Positive Rights	Obligates others to do something on your behalf

If you have a right, others have a duty to not violate that right (stealing, murder, ...)

- Kant argues the opposite: rights follow from duties.

Virtue Ethics (or Character-based ethics)

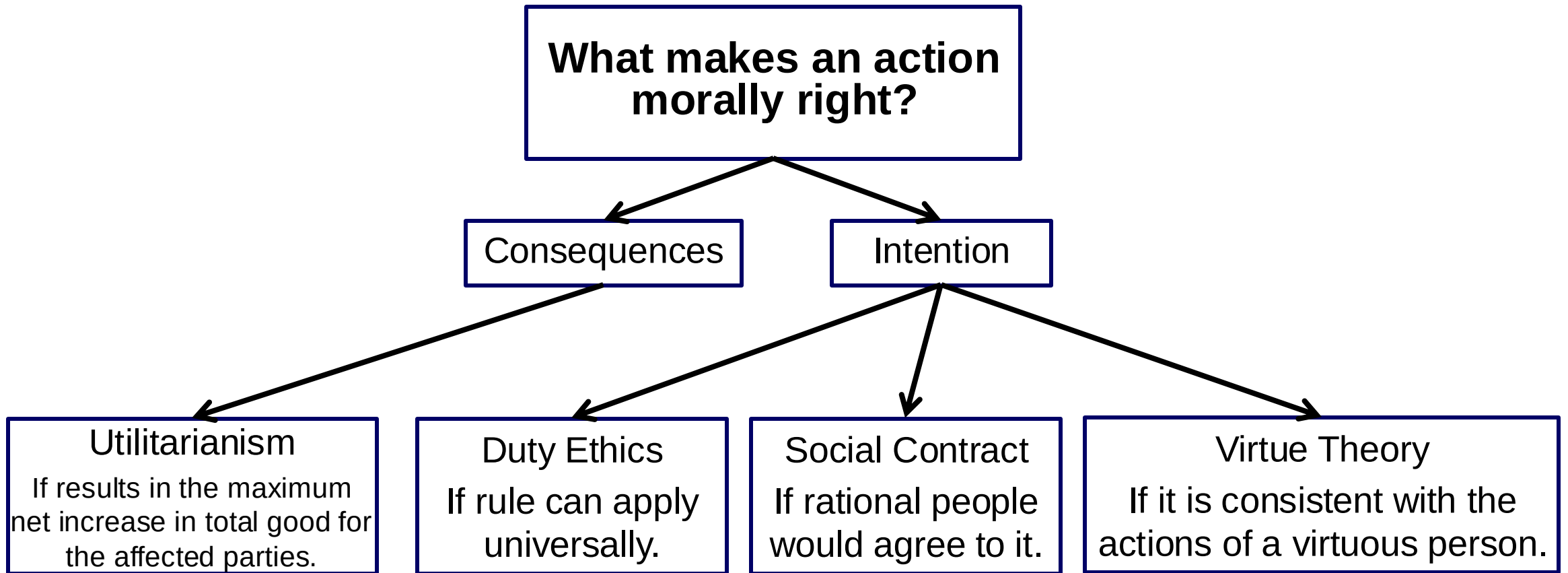
- Based on the philosophy that ethical behavior supports good character traits: honesty, courage, loyalty, competence, friendliness, and responsibility
 - Aristotle, Socrates, Plato
- Person based, not action based.
 - Actions are considered right if they support good character traits (virtues).
 - Applies more broadly to a person's life than a single action.



"We do good things because we are virtuous people and seek to enhance these character traits in ourselves and in others"

Aristotle

Review: Differences in Theories



What is Engineering Ethics?

- **“Engineering ethics ... is the study of the moral values, issues, and decisions involved in engineering practices.”**

Schinzinger and Martin, 2000

- **Governs the choices one makes in practicing engineering.**

Professional Codes of Ethics

- Professional code of ethics includes principles, rules and standards that govern the conduct of engineers in their roles as professionals.
- What a code of ethics isn't ...
 - Not Laws
 - Not Complete Frameworks
 - Not Algorithms
 - Not Exclusive Checklists

Example Codes of Ethics

- **IEEE Code of Ethics**
 - 10-point set of principles and rules that apply to members
 - <https://www.ieee.org/about/corporate/governance/p7-8.html>
- **ACM Code of Ethics and Professional Conduct**
 - More elaborate conceptual framework developed around a member's role: general moral imperatives as a human being; professional responsibilities as a practitioner; organizational leadership imperatives
 - <https://www.acm.org/code-of-ethics>
- **The Software Engineering Code of Ethics and Professional Practice**
 - Developed by IEEE-CS/ACM joint task force
 - 80 very specific rules for software engineers
 - <https://ethics.acm.org/code-of-ethics/software-engineering-code/>

3 Major Themes Covered in These Codes

- 1. General ethics principles: shared human values and societal ideals.**
 - honesty, loyalty, human welfare, respect for privacy, fair treatment, respect for public health, safety and environment.
- 2. Professional responsibilities and duties: rules (sometimes very specific) governing professional activities of members.**
 - professional competence, relevant laws and regulations, confidentiality, conflict of interest, ...
- 3. Leadership and Management Responsibilities:**
 - ACM Code and Software Engineering Code include a specific section on practitioners who serve in leadership roles

IEEE Code of Ethics (1-6)

- I. To uphold the highest standards of integrity, responsible behavior, and ethical conduct in professional activities.**
 - 1. to hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment;**
 - 2. to improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, including intelligent systems;**
 - 3. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;**
 - 4. to avoid unlawful conduct in professional activities, and to reject bribery in all its forms;**
 - 5. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, to be honest and realistic in stating claims or estimates based on available data, and to credit properly the contributions of others;**
 - 6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;**

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Is it against IEEE code of conduct to capture features from one's face as a biomarker and use it to identify them with a camera?

Yes / No

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What if it's used as a form of two-factor authentication to improve privacy and protect a user's data?



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Conflicts of Interest (COI)

- A COI is when you have a certain personal interest that may interfere with the interests and duty you have to your employer.
 - They can be actual, or perceived, and **some argue there is no difference**

Perceived COI Example

Let's imagine a club is raffling a new car. The winner's name is drawn randomly, and it turns out it is the club's general manager. The manager may have bought the ticket like anyone else and won by chance. But people would lose confidence in the integrity of the drawing, or of the club, due to a perceived COI.

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Is it against IEEE code of conduct to develop a program that is designed to break the law?

Yes / No

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Yes / No

What if it's a program to guide a vehicle across a solid, double yellow line to avoid an obstacle?



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Data Falsification – Famous Examples



Enron – massive energy company

Financial audits were hiding \$B's of debt from shareholders

Discovered in 2001, and \$73B of shareholder value was lost

Both founder and CEO were convicted of felonies

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VW – Emissions scandal – aka “Dieselgate”

Engines in 11M cars could detect when in emissions testing, and only activate emissions controls during test.

Discovered in 2015. Prison time, recalls, vouchers.



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Theranos – startup developing blood testing tech

Misled investors and customers on accuracy of testing

Raised ~\$1B of investment; contract with Walgreens

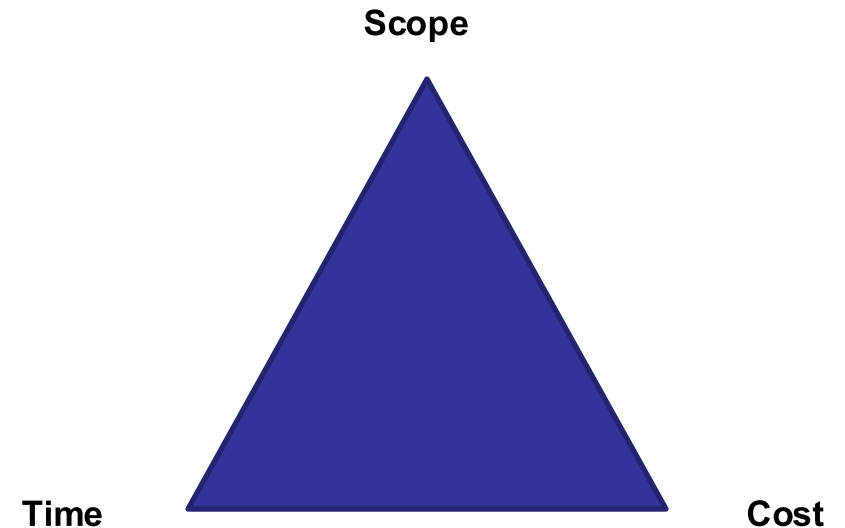
2018 – company dissolved, May 2023 founder in prison

IEEE Code of Ethics (7-10)

- II. To treat all persons fairly and with respect, to not engage in harassment or discrimination, and to avoid injuring others.**
 - 7.** to treat all persons fairly and with respect, and to not engage in discrimination based on characteristics such as race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;
 - 8.** to not engage in harassment of any kind, including sexual harassment or bullying behavior;
 - 9.** to avoid injuring others, their property, reputation, or employment by false or malicious actions, rumors or any other verbal or physical abuses;
- III. To strive to ensure this code is upheld by colleagues and co-workers.**
 - 10.** to support colleagues and co-workers in following this code of ethics, to strive to ensure the code is upheld, and to not retaliate against individuals reporting a violation.

Decision Making in Engineering

- **Decisions are usually guided by development and production variables:**
 - Cost (Money)
 - Time (Schedule)
 - Scope (Features + Quality)
- **A change in one effects the others**
 - Less time with same scope → more money
 - Less money and same time → less scope
 - Usually not a linear relationship



Decision Making Under Pressure

- Can lead to mistakes in
 - estimating time → increased cost or decreased scope
 - estimating scope → increased cost or increased time
- Unexpected bugs → > time, > \$\$, or < scope
- Pressure to maintain schedule, and cost puts pressure on scope (features + quality)
 - which can lead to ethical decisions to release a product too early
- Personal gain also enters the picture (example conflict of interest)
 - What will happen to your reputation, promotion, bonus, ...

Decision Making Under Pressure



Dealing with an Ethical Dilemma

- **Gather Information**
 - Talk to people involved (if possible/appropriate)
- **Determine significance and your ethical responsibility**
 - Are there ways to avoid the ethical issue by taking another path?
 - Examine possible outcomes, your duties, violation of rights and values, ...
- **Communication**
 - Talk to friends and family to get opinions (if appropriate)
- **Keep a physical record of all activities**
 - Phone calls, Emails, personal meetings, ...

Whistleblowing

- A whistleblower is a person that discloses information within an organization that is illegal, unethical, fraud, abuse, etc.

Whistleblower protection

- False Claims Act (1863) [Relates to fraud against the government]
- Sarbanes-Oxley Act (2002) [Outgrowth of Enron scandal]
 - “Whoever knowingly, with the intent to retaliate, takes any action harmful to any person, including interference with the lawful employment or livelihood of any person, for providing to a law enforcement officer any truthful information relating to the commission or possible commission of any federal offense, shall be fined under this title, imprisoned not more than 10 years, or both.”
- General protection from harassment or discrimination

Seek legal advice first...

When to blow the whistle?

- **Need**
 - Clear and important harm that can be avoided
 - Illegal activities you may be pulled into
- **Proximity**
 - You are in a clear position to report problem (not hearsay)
 - Relevant expertise to judge there is a problem
- **As a last resort**
 - Not as a method to exact revenge or for personal gain.

Summary

- **There is no single moral expert or ethical theory that everyone agrees on**
- **Turn to Codes of Conduct as a guide**
- **You have a duty and moral responsibility to act if you observe fraud or illegal activity**
 - **Think UM Honor Code “... nor have I concealed any violation of the Honor Code”**

Next Week

**Topic: AI and Ethics: Trust,
Cooperation, and Society**

Ben Kuipers
Professor Emeritus, EECS



Attendance Answer

La Liga